KAINAT RIAZ

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AI researcher working on uncertainty and introspection in LLM agents — test-time reinforcement learning, epistemic uncertainty, and human-in-the-loop agent decision-making.

EDUCATION

MS EE (AI & Autonomous Systems) — National University of Sciences and Technology (NUST) 2020 – 2022

- GPA 3.90/4.00 — graduated 3rd on merit. Thesis: *Targeted Adversarial Attack on Classic Vision Pipelines* (published in CVIU, 2024).

BS Electrical Engineering (Electronics) — Air University 2016 – 2020

- GPA 3.42/4.00 — merit scholar with merit-based tuition fee waiver.

PUBLICATIONS & MANUSCRIPTS

Under review

- Riaz, K.*, Mohsin, M.A.*, Bilal, A., Umer, M., Mohsin, A., Riaz, A., Subhan, A., Cioffi, J.M. *"Epistemic Uncertainty for Test-Time Discovery."* arXiv preprint arXiv:2605.11328, under review at ICLR 2027. (*equal contribution; collaboration with Stanford University)

Published

- Riaz, K., Anjum, M.L., Hussain, W., Manzoor, R. *"Targeted Adversarial Attack on Classic Vision Pipelines."* Computer Vision and Image Understanding, Vol. 249, Art. 104140, December 2024.

RESEARCH EXPERIENCE

Independent Researcher — LLM Agents & Test-Time Learning — in collaboration with Stanford University 2025 – Present

- Co-first author of *Epistemic Uncertainty for Test-Time Discovery* (arXiv:2605.11328, under review at ICLR 2027), joint work with Prof. John M. Cioffi's group at Stanford University.
- Designed a mutual-information-shaped advantage with nuclear-norm regularization for test-time RL that sustains ~2.2× higher solution diversity than the clipped-IS baseline and raises maximum reward on 3 of 4 scientific-discovery benchmarks (Erdős minimum overlap, autocorrelation inequalities, GPU kernel optimization).
- Ran large-scale multi-node RL experiments (SLURM, up to 120B-parameter models) spanning mathematics, GPU-kernel, algorithm-engineering, and computational-biology benchmarks; designed and executed ablation and robustness studies for peer review.
- Work in progress — *when-to-ask*: investigating whether cross-trajectory divergence of mid-layer hidden states predicts when a long-horizon coding agent should ask for human input, earlier and more accurately than output-divergence baselines; built the N-trajectory white-box harness, asynchronous hidden-state alignment, and AUROC/lead-time evaluation suite.

Research Assistant — Robotics & Machine Intelligence Lab, NUST 02/2022 – 01/2023

- Led research on combating adversarial attacks through experimentation with state-of-the-art GANs, deep-learning denoisers, and blind deconvolution methods.
- Designed targeted and deep black-box adversarial attacks on vision systems built with handcrafted features; performed statistical, qualitative, and quantitative analysis across robotics and computer-vision pipelines.
- Designed an automatic loop-closure detector evaluated on the KITTI dataset; prepared materials for peer-reviewed journal submissions.

WORK EXPERIENCE

Staff Software Consultant (Agentic AI) — 10Pearls 06/2026 – Present

- Own the semantic/ontology layer and the agent runtime of AiX, an air-gapped, Palantir Foundry-class operational platform delivered for Blackford Technologies (UAE); authored 20+ architecture decision records across both layers.
- Raised agent accuracy from 7/12 to 12/12 on three consecutive runs of a 12-question benchmark over an OSDU subsurface dataset (95,260 objects, 243,957 edges) by adding a deterministic multi-hop path planner and precise model-facing tool descriptions, clearing the two-hop reverse-traversal queries that had been the ceiling.
- Eliminated the runtime's worst failure mode, confabulated success: the agent now confirms writes by read-back instead of blind retry, receives real tool errors rather than sanitised ones, and distinguishes an invalid request from an upstream outage.

AI Engineer — Wanclouds Inc. 09/2023 – 05/2026

- Developed an autonomous AI agent for cloud-infrastructure monitoring, troubleshooting, and automated issue resolution across multi-cloud environments, reducing downtime and manual intervention.
- Built a unit-testing framework for AI agents using DeepEval, integrated into GitHub workflows to automate validation; implemented a DSPy prompt optimizer to cut prompt-engineering time.
- Designed a proprietary dataset and fine-tuned a small LLM on the company codebase, creating a privacy-compliant code assistant; reduced model training time from 2.5 hours to 10–20 minutes.
- Conducted adversarial evaluations with vulnerability scanners to patch security weaknesses in deployed LLMs; optimized fine-tuned LLMs for deployment on smaller GPUs and CPUs.

AI Consultant — Macknowsoft 08/2024 – 01/2025

- Single-handedly delivered a production-ready Arabic-language RAG chatbot for a Saudi client: Elasticsearch migration, embedding-based semantic search, citation-grounded answers, and the full backend (APIs, authentication, PostgreSQL).

Machine Learning Engineer — NetSol Technologies 01/2023 – 09/2023

- Fine-tuned Stable Diffusion (DreamBooth, textual inversion) for consistent storybook character design; implemented ControlNet for a furniture-design mockup workflow.
- Trained YOLOv8 on a custom dataset for automated retail inventory tracking; built a hybrid CNN/rule-based classifier achieving 96% accuracy in radar-based patient fall detection.
- Led RFP development for a Defense Ministry Green-AI project proposing AI-driven agricultural solutions.

HONOURS & AWARDS

- 3rd Merit Position, MS EE (AI & Autonomous Systems) — NUST, 2022.
- Merit Scholar (top undergraduate performer, tuition fee waiver) — Air University, 2019.
- Certificate of Excellent Result, FSc Pre-Engineering — Army Public School and Colleges, 2016.

SKILLS

- **ML / LLMs:** PyTorch, TensorFlow, QLoRA/PEFT, bitsandbytes, DSPy, DeepEval, Hugging Face, OpenCV, scikit-learn, vector stores, SLURM multi-node training, LangGraph, agent tool contracts, agent evaluation harnesses.
- **Engineering:** Python, FastAPI, SQL (MySQL, PostgreSQL), Elasticsearch, Git, Linux, NumPy, Pandas, SciPy, Matplotlib, Seaborn, Pydantic, SQLAlchemy/Alembic, pgvector/Qdrant, Docker, Kubernetes.

CERTIFICATES & LANGUAGE

- Generative AI with Large Language Models — Coursera.
- English: IELTS 7.5.